Cesar Rodriguez

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EDUCATION

University of Michigan **May 2023, US, MI Ann Arbor**
Bachelor's of Science in Engineering in Computer Science - Magna Cum Laude **3.55/4 GPA**
Community Involvement: Michigan Artificial Intelligence Safety Initiative, Society of Hispanic Professional Engineers
Relevant courses:
- Engineering: Machine Learning, Introduction to Artificial Intelligence, Web Systems, Human Centered Software, Introduction to Logic Design, Data Structures and Algorithms, Introduction to Computer Organization, Introduction to Computer Security
- Philosophy: Mind and Machine, Descartes to Kant, Early Modern Philosophy, Critical Reasoning
- Statistics: Statistics and Artificial Intelligence, Computational Methods, Introduction to Statistical Computing

EXPERIENCE

NextArena **US, IL, Chicago**
Chief Executive Officer **July 2026 - Present**
- Lead a team of three engineers and one graphic designer developing an esports tournament platform, launched October 2026.
- Developed an automated outreach and email system for player announcements and for partnership outreach to content creators, esports lounges and collegiate esports teams.

Full Stack Developer - Technical Lead **January 2023 - July 2026**
- Implemented the company website in accordance with branding specifications, collaborating with designers to deliver an intuitive user experience.
- Developed and documented frontend and backend REST APIs and database architecture to optimize data flow and improve performance.
- Implemented a payment system supporting cryptocurrency and fiat transactions using row-level database locking for concurrency control, a state machine to enforce valid transaction transitions, and an append-only ledger for auditable transaction history.

MIT Lincoln Laboratory **US, MA, Westford**
Space Systems Analysis and Test Intern - Group 95 **May 2023 - September 2023**
- Developed a distributed PXE boot server using TCP, HTTP, and TFTP to automate Red Hat Enterprise Linux 8 deployments through Bash, Python, and C++.
- Reduced backend deploy times by 95% or more, and reduced setup complexity by 50% by testing on virtual machines and exporting to sensors.
- Implemented hardware-in-loop (HIL) testing automation strategy to analyze performance, reliability, extensibility, and long-term success.

Intel Corporation **US, CA, Folsom**
Software Engineering Intern - Cohort Lead **May 2022 - August 2022**
- Created a plugin for an internal tool enabling integration of Intel's new security engine in Intel Firmware Images, improving firmware quality.
- Ensured compatibility with external firmware images by developing Python and Bash tooling that standardized image generation.
- Refactored C++ class hierarchies to improve maintainability and interoperability with the new firmware engine.
- Conducted code reviews and utilized pair programming to develop and implement new features for the new firmware engine.

Intel Corporation **US, IL, Chicago**
Software Engineering Intern - Cohort Lead **May 2021 - August 2021**
- Standardized stress testing operating system for validating 14th generation central processor units, resulting in increased performance and reliability.
- Modernized legacy Python tooling across multiple teams, earning four Intel recognitions and reducing execution time by approximately 4%.
- Developed and evaluated simulated models to test CPU functionality on FPGA, resulting in a report of findings and recommendations to the team.
- Trained five international team members in Malaysia on usage and implementation of FPGA simulated model for testing CPU functionality.

PROJECTS

University of Illinois Chicago **US, IL, Chicago**
Freelance Data Analyst **December 2017 - May 2022**
- Extracted unstructured data from image sources to produce human-readable dataset for use by non-technical Museum Studies Program faculty.
- Cleaned, structured, and analyzed datasets supporting research into government accountability.
- Designed data structures to promote ease of use, scalability, and accessibility on a website accessed by tens of thousands of users.

University of Michigan **US, MI, Ann Arbor**
Research Assistant, Cardiovascular Technology **October 2019 - May 2020**
- Interviewed healthcare professionals to identify workflow bottlenecks in emergency cardiovascular response and inform application design.
- Created an application to improve communication between medical personnel that reduced response time from 7 minutes to 5.5 minutes by automating note-taking and streamlining the communication process.

SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, SQL, Java, R, MATLAB, Verilog

Frameworks: React, Django

Technologies & Tools: Node.js, Git, Linux, Visual Studio, Cadence Perspec, Arduino, AutoCAD, Photoshop

Operating Systems: Windows, Ubuntu, Red Hat Enterprise Linux, macOS